MILITARY SPECIFICATION

HEATER, DUCT TYPE, PORTABLE, TRAILER MOUNTED HDU-13/M

"This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense."

1. SCOPE

1.1 This specification covers a trailer-mounted, 100,000 BTU/Hr, portable, duct-type heater designed for safe operations in atmospheres containing explosive mixtures of aviation gasoline or jet fuels and air, designated HDU-13/M.

1.2 Classification. The heater shall be of the following classes, as specified (see 6.2):

Class 1 - Equipped with a 220-440 volt (v), 3 phase, 60 hertz (hz) electric motor.

Class 2 - Equipped with a 200v, 3 phase, 400 Hz electric motor.

1.2.1 Part number. Specification part number for items described in this specification will be formulated as shown in Section 6.5.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Federal

CC-M-1807	Motor, Alternating Current, Fractional and Integral Horsepower (500HP and smaller)
PPP-T-60	Tape; Packaging, Water Proof

Military

MIL-B-121	Barrier Material, Grease Proofed, Water Proofed Flexible

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDO, Robins AFB GA 31098 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

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MIL-V-173	Varnish, Moisture and Fungus-Resistance (for treatment of communications, electronic and associated equipment)
MIL-T-704	Treatment and Painting of Materiel
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-C-5756	Cable and Wire, Power, Electric, Portable
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-19834	Plates, Identification or Instruction, Metal Foil Adhesive Backed General Specification for
MIL-B-26195	Box, Wood-Cleated, Skidded, Load Bearing Base
MIL-D-38386	Duct Assemblies, Ground, Conditioned Air, Insulated Flexible

STANDARDS

Federal

FED-STD-595	Colors

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of

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MIL-STD-463	Definitions and System of Units, Electromagnetic Interference Technology
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIl-STD-882	System Safety Program, for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS 25486	Connector, Plug, Attachable, External Electric Power, Aircraft, 115/200 volt, 400 hertz
MS 27265	Connector, Plug, Electrical-Ramp Power, 416/ 240 volts, male

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

National Bureau of Standards Handbook H-28

Screw Thread Standards for Federal Services

(Application for copies should be addressed to the General Services Administration, Specifications Branch, Bld 197, Washington Navy Yard, Wash DC 20407)

Department of the Labor, Occupational Safety and Health Administration

Occupational Safety and Health Act (OSHA) with Supplemental Revisions Title 29, Code of Federal Regulations, Chap XVII, Part 1910 and Amendments.

(Applications for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington DC 20402)

DRAWINGS

Air Force

7545352	Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground and Ground Support Equipment
7737593	Air Force Change Drawings - Protective Finish, Ground and Ground Support Equipment

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society of Mechanical Engineers (ASME)

Power Test Code

19.5 Interim Supplement 19.5 on Instruments and Apparatus; Application, Part II of Fluid Meters

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York NY 10017.)

National Electrical Manufacturer's Association Standard

MG-1 Standards for Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturer's Association, Engineering Department 155 East 44th Street New York NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE BULLETINS

S1.4 Specification for Sound Level Meters

S1.6 Preferred Frequencies and Band Numbers for Accoustical Measurements

S1.11 Specification for Octave, Half-Octave, and Third-Octave Band Filter Sets

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York NY 10018.)

NATIONAL FIRE CODES

Volume 6 National Electrical Code (NFPA No. 70)

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston MA 02110)

Underwriters Laboratories Inc.

UL 674(B) Electric Motors and Generators for use in Hazardous Locations, Class 1, Groups C and D

(Copies of UL Standards may be obtained from the UL Publications Stock, 333 Pfingsten Road, North Brook, IL 60062.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. The specification makes provision for preproduction testing.

3.2 Safety. The heater shall be provided with safety features and devices for the protection of personnel and equipment. Parts which are hazardous to personnel shall be guarded. All equipment furnished under this specification shall be analyzed for Category I and Category II hazards as defined in MIL-STD-882. Equipment must be consistent with applicable OSHA standards. Any hazard so identified shall be brought to the attention of the procuring agency with recommended action for satisfying safety requirements. Action order of precedence shall be as specified in MIL-STD-882, Para 5.4.2.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment. When materials are used in the construction of the heater that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with performance requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Recycled materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. None of the above shall be interpreted to mean that the use of used or rebuilt products will be allowed.

3.5 Design and construction. The heater shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Reliability. The heater shall have a mean-time-between-failures (MTBF) of not less than 434 hours at 0.9 confidence (see 6.4.1 and 6.4.2).

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3.5.2 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided, where practicable. That mounting shall permit the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.3 Explosion-proof requirements. The heater shall be of explosion-proof design suitable for use in class I, group D atmospheres and class I, division 1 hazardous locations in accordance with NFPA No. 70.

3.5.4 Air temperature differences. The heater shall be so designed that the maximum air temperature difference will be less than 15°F between any two locations in a duct at a distance of 10 inches from the heater discharge.

3.5.5 Air transportability. The heater shall be constructed to meet the requirements of MIL-A-8421.

3.5.6 Mobility. The trailer-mounted heater shall be designed to meet the mobility requirements of MIL-M-8090, Type II, Group C. Only the following requirements listed in 3.2 of MIL-M-8090 shall apply:

a. General Design

b. 3.5.1.1 Steering Geometry, Group C

c. Parking Brakes

d. Pintle Hook, Rear

e. Tires and Tubes

f. Towbars and Lunette Eyes

g. Wheels

h. Markings. 3.19 and 3.19.3 shall not apply.

3.6 Performance. The heater shall withstand the following conditions without detrimental effect to subsequent operation:

a. Operating temperatures ranging from -65°F to +125°F and exposure to temperatures ranging from -80°F to +160°F.

b. Altitudes ranging from sea level to 5,000 feet.

c. Rainfall as encountered in any locale.

d. Fungus growth as encountered in tropical climates.

e. Exposure to salt-fog.

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3.6.1 Delivery. At standard atmospheric pressure, the heater shall deliver uncontaminated, heated air at the following rates when the discharge temperature, pressure, and flow rate are measured at the end of a 15-foot section of insulated duct while using only one of the two heater outlets.

3.6.1.1 Rated output. At -20°F ambient temperature, the heater shall continuously deliver at least 39 pounds of air per minute (pm) of 155°F air against static pressures ranging from 0 to 12 inches of water.

3.6.1.2 Variable output. The heater shall deliver heated air when operating in any ambient temperature between -65° and +100°F. The unit shall fulfill the following airflow requirements while operating against static pressures ranging from 0 to 12 inches of water:

a. At an ambient temperature of -20°F, the unit shall deliver air from 39 to 44 pm at any selected discharge temperature between 80° and 150°F.

b. At an ambient temperature of 0°F, the unit shall deliver air from 37 to 42.5 pm at any selected discharge temperature from 100° to 185°F.

c. At an ambient temperature of 70°F db and 65° wb, the unit shall deliver 36 pm at any selected discharge temperature 100° to 200°F.

3.6.1.2.1 The heater shall operate as a blower at an ambient temperature from -65° to 125°F. In addition, while operating as a blower, the heating units shall not operate but the unit shall deliver at least 45 pm at ambient temperature between -65°F and +70°F against static pressure of 0 to 12 inches of water.

3.6.2 Low-temperature starting. The motor and heat generator shall immediately start without preheating after exposure to an ambient temperature of -65°F as specified in 4.6.14.1.

3.6.3 Operation in tilted position. The heater shall function satisfactorily when its normal operating plane is at an angle of 8-1/2° in any direction from true horizontal.

3.6.4 Operating life. The heater shall be capable of operating for a minimum of 1,000 hours without failure of any major part.

3.6.5 Dielectric strength. The insulation of each electrical circuit of the heater shall withstand for a period of a 1-minute, without breakdown, the application of a 60-cycle potential of twice the normal voltage plus 1,000V (rms).

3.7 Details of components

3.7.1 Heat exchanger assembly. The heat exchanger assembly shall be constructed of aluminum, copper or other nonsparking material and shall be designed to obtain the highest efficiency practical. The assembly shall be designed to withstand the most severe explosion conditions that can occur with any fuel mixture without adversely affecting heater performance. The assembly shall be designed to permit disassembly and cleaning of all parts.

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3.7.2 Heat generator assembly. The heat generator shall consist of immersion type electric resistance elements suitable for operation under all conditions specified herein. Heating elements shall be immersed in the secondary heat transfer fluid and contained within an explosion-proof, closed-system, low pressure housing.

3.7.2.1 Secondary fluid system. The heater system shall utilize undiluted diethylene glycol as a secondary heat transfer fluid. This fluid shall be circulated by a centrifugal pump through appropriate plumbing from the heat generator to the heat exchanger and return. An explosion tank, filler port, drain valve, and pressure relief valve shall be included in the system.

3.7.3 Hotspots. Exposed surfaces of the heat exchanger, heat generator, motors, control boxes, or conduit fittings shall not exceed $325^{\circ} \pm 5^{\circ}$ under any operating conditions. The surface temperatures of these components shall be monitored by an adequate number of firmly attached thermocouple leads which will convey surface temperature indications to the control box. Circuit breaker(s) located within the control box shall be responsive to the thermocouple temperature indications and shall break the heat generator power supply circuit if the surface temperature reaches $325^{\circ} \pm 5^{\circ}F$. Thermocouple wires need not be installed in accordance with NFPA No. 70, except to maintain explosion-proof integrity of the control box.

3.7.4 Heat control system. The heat control system shall provide for automatically maintaining any desired average discharge air temperature between 40° and $200^{\circ}F$ within the limits specified in 3.6.1.2 regardless of change in airflow rates. A control knob or lever shall be located on the instrument panel for manual selection of the desired discharge air temperature. This control shall include a 360° dial and pointer to show percentage of control travel and motion required for increase or decrease of the outlet temperature. The heat control system shall automatically maintain the desired outlet temperature within $\pm 10^{\circ}F$ when once established.

3.7.4.1 Temperature indicator. A temperature indicator and a remote sensing element capable of indicating the average heater discharge air temperature within $\pm 10^{\circ}F$ shall be provided. The temperature indicator shall be calibrated for the operating range of the heater. The portion of the indicator dial face below the $40^{\circ}F$ mark shall be green, the portion above the $200^{\circ}F$ shall be red; and the portion between 40° and 200° shall be white. The temperature indicator shall be independent of the operating controls.

3.7.4.2 High-temperature limiting devices. High temperature limiting devices shall be provided to prevent overheating in the heated airstream and within the secondary heat-transfer system used in the unit. The devices shall be automatically actuated, but manually reset. The high-temperature limiting devices shall actuate to break the heat generator power supply circuit when the airstream temperatures reach $215^{\circ} \pm 5^{\circ}F$ or when the secondary fluid temperature reaches its maximum design temperature of $325^{\circ} \pm 5^{\circ}F$.

3.7.4.3 Operating light. An operating light shall be included in the power circuit that will glow only when the heating coils are drawing current. The lens color shall be green, at least 1/2 inch in diameter, and visible to the operator at a distance of 100 feet from all four sides.

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3.7.5 Air blower. The air blower shall supply the quantity of air specified herein when operating against static pressure of 0 to 12 inches of water measured at the end of one 15-foot flexible duct. Aluminum or other nonsparking material shall be used for construction of the blower, wheels, hubs, and scrolls. It shall be possible to operate the air blower as a ventilating blower with the heating system inoperative as specified in 3.6.1.2.1.

3.7.5.1 Ventilating airflow control. The rate of ventilating airflow shall be controlled by a manually operated restricting shutter or shutters. The rate of airflow shall be variable from 25 percent to maximum.

3.7.5.2 Automatic overload control. If the power necessary for the capacity specified in 3.6.1.2 is exceeded during low static pressure operating of the heater, an automatic control shall be incorporated at the blower to prevent overloading of the power supply circuit.

3.7.5.3 Ducts. The heater shall be provided with two 8-inch nominal diameter insulated air duct assemblies each 15 feet in length which are in accordance with MIL-D-38386. The duct assemblies shall include connectors as specified in MIL-D-38386 except that the finish shall be zinc or chrome plating for spark resistance as specified in 3.13. A system shall be incorporated into the ducts which will provide a continuous ground from one end to the other terminating in a grounding strap at the far end. Unless otherwise specified, the duct color shall be green, color number 24052 per FED-STD-595.

3.7.6 Power package. The power package shall contain all necessary components for proper operation of the heater from an external 3 phase, 4 wire electrical power source.

a. Class 1 Heater. 220/440V ± 10%, 60Hz. The total current draw of the heater shall not exceed 104 amp under 200V operation and 52 amp under 440V operation. When delivered, the unit shall be connected for 440V operation.

b. Class 2 Heater. 200V ± 10%, 400Hz. The total current draw of the heater shall not exceed 104 amp.

3.7.6.1 Electric motor. The motor shall be rated for continuous duty at, as above, the most severe power requirement of the heater at any ambient temperature within the operating range. Motor insulation shall be Class H suitable for the intended application. Temperature rise at this rating shall be in accordance with NEMA standard MG-1.

a. Class 1 Heater. The electric motor shall be a 220/440V, 3 phase, 60 Hz, squirrel cage induction type conforming to CC-M-1807, Type II, Class 3, Style E with Enclosure 2, Design L.

b. Class 2 Heater. The electric motor shall be a 220V, 3 phase, 440Hz, squirrel cage induction type conforming to UL 674(B), Class I, Group D.

3.7.6.1.1 Starter relay or relays. A polyphase, alternating current, across-the-line starter relay or relays shall be provided with under voltage and overload protection.

3.7.7 Wire and cables. All wiring and cables used in the electrical circuit shall conform to the National Electrical Code, NFPA No. 70, Article 330, for mineral-insulated (MI) cable. Wiring shall be firmly supported to

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prevent undue stress on the conductor terminals and to eliminate the possibility of a shift in conductor position that may result in a degradation of the performance specified herein. All wiring shall be installed in accordance with Article 500 and 501 of NFPA No. 70. Clearance between wiring and heat-generating parts shall be sufficient to prevent damage to wire insulation, sufficient slack shall be provided to prevent stress on cable forms, wires, terminal connections, and to facilitate replacement of parts without disconnecting other parts.

3.7.7.1 Number coding. All wiring shall be number-coded. Wire under 1/4-inch overall diameter shall be coded on both sides diametrically opposite at intervals varying from 1 to 6 inches center-to-center. Large wires shall be marked by cable markers at each detachable end.

3.7.7.2 Terminations. Before being soldered to terminals, wires shall be effectively secured to the terminals so that the connectors are not dependent on soldering alone for mechanical strength. Varnish or lacquer conforming to MIL-V-173 shall be applied to the ends of fibrous coverings, binders, and similar items when not secured against fraying by other means.

3.7.7.3 Terminals. Terminals attached to wires may be of either the solder or solderless type, except for the following limitations on the solderless type.

3.7.7.3.1 Solderless terminals. The solderless terminals shall be of a type that has means for gripping the wire insulation. The terminals shall be used only on stranded conductors ranging from 642.6 to 10380 circular mils (No. 22 to 10AWG).

3.7.7.4 Relay adjustment. All relays shall operate without chatter or hum and shall hold settings for long periods of service. The relays shall operate under the service conditions specified herein without readjusting the contact spring tension adjustment. Where a relay is required to go from operate to nonoperate positions with small current or voltage differentials, the compensation for temperature effects shall be accomplished by externally connected controls.

3.7.7.5 Grounding surfaces. Where a nonconductive protective coating, such as paint or enamel, is used on the equipment, conduit, couplings, or fittings, such coating shall be removed from threads and other contact surfaces to insure a good electrical connection.

3.7.7.5.1 The unit grounding wire shall be a 25-foot long, extra flexible, tinned copper cable with end clip having less than 25 ohms resistance to ground from the extreme end of two flexible ducts connected in series to unit, through unit and ground cable.

3.7.7.6 Power cable. The power lead-in cable shall be a 50-foot long, extra-flexible, 6-conductor, 600V cable conforming to MIL-C-5756. The wire size shall be selected to carry the full load current at 220V.

a. Class 1 Heater. The cable shall contain 6 conductors, 4 primary and 2 for the control circuit. A connector conforming to MS 27265 shall be molded to the cable.

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b. Class 2 Heater. The cable shall contain 4 conductors. Control circuit conductors are not required for the Class 2 heaters. A connector conforming to MS 25486 shall be molded to the cable.

3.7.7.6.1 Cable compartment. A compartment shall be provided within the heater to provide for storage of the power lead-in cable when it is not in use.

3.7.7.6.2 Rack. A rack for the electric power cables shall be permanently fastened to the interior of the cable compartment.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.9 Electromagnetic interference. The electrical components that are to be used as an integral part of the heater shall be so designed and assembled using the latest suppression techniques that the complete unit will meet the radiated and conducted interference requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463 for Class II B equipment.

3.10 Noise level. The noise level generated by the heater shall not exceed the values specified in Table I when measured as specified in 4.6.13.

TABLE I

Noise Level Requirement

Octave Band Center Frequencies (CPS)	Octave Band Sound Pressure Levels (In Decibels Ref. 0.0002 Microbar)		
	25-Foot Circle		
	Operator	Maximum	Average
63	78	68	67
125	85	75	72
250	86	76	72
500	85	75	70
1000	83	73	68
2000	76	66	61
4000	71	61	55
8000	66	56	50

3.11 Dimensions. The dimensions of the heater shall not exceed 42 inches in width, 48 inches in height, and 70 inches in length, excluding tow bar provisions, pintle hook, and removable tiedowns.

3.12 Weight. The dry weight of the entire assembly shall not exceed 1,300 pounds.

3.13 Finishes and protective coatings. All finishes and protective coatings on the heater shall be in accordance with MIL-STD-808 except as amended by the US Air Force Drawings 7545352 and 7737593.

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3.13.1 Color. The color of the heater shall be green, color 24052, conforming to FED-STD-595. This shall apply to all parts that are visible externally except lubrication fittings, tiedown rings, or other plated parts. Internal components need not conform to this color.

3.13.2 Nonsparking finishes. For spark-prevention purposes, exposed ferrous parts, such as tiedown rings, pintle hook assembly, wheel hubs and rims, shall be zinc- or chrome-plated as specified in 3.13.

3.14 Operational markings

3.14.1 Operating instructions. Brief instructions for starting, operating, and shutdown of the heater shall be permanently and conviently located on or adjacent to the heater control panel. The instructions shall be such that oil, dirt, water, or other contaminents, will not affect their legibility nor cause them to be erased.

3.14.2 Warning plate. A reversible warning plate shall be fastened to the outside of the cable compartment door indicating appropriate voltage (220/440) at which the unit is to be operated. One side of the plate shall contain a 220V wiring diagram with instructions for changeover to 440V and the other side shall have a 440V diagram with instructions for changeover to 220V. This warning plate is not required on the Class 2 heaters.

3.14.3 Trailer marking

3.14.3.1 Marking for hoisting. The following marking shall be located near the hoisting provisions in block type, 1-inch high, black letters:

USE CABLE GUARDS WHEN HOISTING.

3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following additional information shall be included:

Special characteristics:

Voltage_____
Phase_____
Frequency_____

Serial No._____

3.16 Workmanship. The heater, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.16.1 Dimensional tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the heater, they shall be held or limited accordingly.

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3.16.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight shall mean that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.16.3 Riveting. Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

3.16.4 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities.

3.16.5 Cleaning. All parts of the heater shall be thoroughly cleaned of loose, spattered, or excess solder or welding material, metal chips, grease or oils, and other foreign material prior to application of protective coatings. Burrs and sharp edges shall be removed.

3.17 Instrument and control panel. The heater shall be provided with a composite instrument and control panel on which all gages, switches, warning lights, and other control items shall be mounted. All instruments and controls shall be readily visible and accessible to operating personnel during normal operation of the heater without the necessity of opening a main access door. All items on the panel shall be permanently and legibly marked. The panel shall be protected by a hinged cover. All gages, switches, warning lights and other control items shall be waterproof. The panel shall be illuminated by a manually operated white light.

3.18 Cabinet assembly. The heater shall be enclosed in a lightweight metal cabinet assembly mounted on an integral four-wheel chassis having sufficient strength and rigidity to meet the mobility requirements specified herein. Removable panels shall be provided in the cabinet for access to major components for replacement and overhaul. Doors shall be provided in the cabinet for access to major components and working parts requiring maintenance and servicing operations as well as for access to controls, control panels, duct storage compartment, and air delivery openings.

3.18.1 Weather resistance. The cabinet shall resist weather and prevent rain, water, snow, ice, sand, mud, or dust from collecting inside in quantities that would prevent satisfactory operation of the heater. All cabinet exterior openings (access doors, control panel covers, intake and discharge openings) that may be in an open position during heater operation shall have means for protecting such openings from weather when the heater is not in operation.

3.18.2 Doors. Wherever possible, all doors shall be designed to present a near-flush exterior with the cabinet when the doors are in the open position. Doors hinged at the top, that may fall and cause injury to personnel, shall be avoided. All doors shall have latches for holding them securely in the closed position. Doors shall be equipped with fixtures that will hold the doors firmly open so that strong wind blasts will not cause the doors to close or injure personnel. Door handles shall be flush as possible to provide a neat appearance and shall be convenient for use by an operator wearing arctic gloves.

3.18.3 Insulation. All heater components subject to high temperatures shall be insulated where necessary, to prevent exterior surface temperatures from exceeding 140°F at an ambient temperature of 70°F.

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3.18.4 Intake opening. The heater shall be provided with a single opening for the air entering the ventilation blower. The intake opening shall incorporate a male coupling for attachment of the ducts specified herein and filter provisions.

3.18.4.1 Air filters. All air intake openings in the heater shall be protected by means of heavy wire or expanded metal screens of 1/2 to 3/4 inch mesh to prevent the entrance of large foreign objects. Permanent-type washable filters shall be provided to further screen dust and dirt particles from intake air supplies. The filters shall be nominal 2 inch dry type suitable for filtering flammable atmospheres. A minimum clearance of 2 inches shall be provided between the screen and filters.

3.18.5 Discharge openings. The cabinet shall have two air delivery openings incorporating male couplings for attachment of the ducts specified herein. Delivery of the total amount of air shall be possible through either outlet or both. One cover or manually operated shutter shall be provided for blanking off one outlet for single-duct operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the heater shall be classified as follows:

a. Preproduction testing . . . . . .See 4.4

b. Acceptance tests . . . . . .See 4.5

4.3 Test conditions

4.3.1 Revolutions per minute. Rotational speed shall be determined by means of directly coupled, positive counter that will actually count the revolutions for a period of not less than 1 minute. A combination unit that counts the revolutions while measuring the time required in connection with other data is preferred. Strobotachometers may be used provided calibration data can be furnished to attest to their accuracy. Hand tachometers shall not be used.

4.3.2 Airflow. Airflow shall be measured by means of smooth approach nozzles or flat plate orifices as specified in ASME Code 19.5. Other apparatus, such as pitot tubes and rotameters, may be used only when specifically approved by the procuring activity. The rate of airflow shall be expressed in terms of pounds per minute.

4.3.3 Barometric pressure. Barometric pressure shall be measured by means of a mercurial barometer at the same elevation as the test. The pressure shall be corrected for the temperature of the scale and mercury. Readings obtained from aneroid barometers will not be acceptable.

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4.3.4 Temperature. Temperatures shall be measured by means of appropriately located thermocouples and properly calibrated potentiometers. Insofar as practicable, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit. Laboratory-type mercury thermometers will be permissible if used correctly in the application.

4.3.5 Pressure. Insofar as practicable, pressure and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Gages shall be laboratory type having a dial diameter of not less than 4 inches.

4.3.6 Readings of all pertinent pressures, temperatures, rotational speed, and other data necessary to determine compliance with the requirements specified herein shall be recorded at intervals not exceeding 30 minutes during all tests.

4.3.7 Preliminary run-in. Any preliminary run-in specified by the manufacturer shall be performed prior to submission of the heater for testing. A certified log of all preliminary running time shall be included in the preproduction test report.

4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. The test samples shall consist of two complete heaters, one separate heat exchanger, and one separate heat generator. The two heaters shall be subjected to the tests specified in 4.4.3, the heat exchanger to the test specified in 4.4.3.1, and the heat generator to the test specified in 4.4.3.2.

4.4.2 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure of irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made to production items or proposed to be made. The summary shall also include an analysis of the predicted effectiveness of such corrections.

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

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f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Production tests. The test samples shall be subjected to the following tests as described under 4.6:

Sample No. 1	Sample No. 2
Examination of product	Examination of product
Operational tests	Operational tests
Capacity	Capacity
Overheat controls operation	Overheat controls operation
Air transportability	Mobility
Endurance and capacity	Tilted position
Electromagnetic interference	Dialectric
Noise level	Grounding
Final capacity	Continuity
	Duct tests
	Environmental tests
	Accessibility
	Explosion

4.4.3.1 The heat exchanger shall be subjected to the explosion test specified in 4.6.17.1.

4.4.3.2 The heat generator shall be subjected to the explosion test specified in 4.6.17.2.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling plan and tests . . . . . See 4.5.2

4.5.1 Individual tests. Each heater shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Operational tests . . . . . See 4.6.2

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c. Grounding . . . . . See 4.6.3

d. Continuity . . . . . See 4.6.4

4.5.2 Sampling plan and test

4.5.2.1 Periodic sampling. Heaters that have been manufactured under essentially the same conditions shall be selected as specified in Table II and subjected to the following tests:

a. Maximum capacity . . . . . See 4.6.5

b. Overheat control operation . . . . . See 4.6.6

c. Final inspection . . . . . See 4.6.7

TABLE II

Selection of Samples

Production Quantity	Periodic Sample Size
From first 10 items manufactured	1
From next 50 items manufactured	1
From each succeeding 100 items manufactured	1

4.5.2.2 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or produced later shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.2.1 Individual tests may continue. For production reasons, individual test or other sampling plans may be continued pending the investigation of a sampling tests failure. But final acceptance of the items on hand or items produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The heater shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, finishes and coatings, and satisfactory assembly.

4.6.2 Operational tests. The heater shall be started and operated for a minimum of 1 hour to demonstrate satisfactory operation of all components,

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including the temperature control, overheat control, and interlock controls. All doors shall be examined for satisfactory operation. The trailer shall be checked to determine that the brakes and steering mechanism operate satisfactorily.

4.6.3 Grounding. Components of the heater shall be tested to determine compliance with 3.7.7.5.1. Probes shall be successively applied to bare surfaces on all components and to ground. System continuity shall be such as to prevent buildup of static electric charges in any components.

4.6.4 Continuity. Prior to sealing conduits, wiring shall be checked for electrical continuity. There shall be no indications of shorts between insulated wires and wiring conduits, junction boxes, raceways, et cetera.

4.6.5 Maximum capacity. The maximum capacity of the unit at the prevailing ambient temperature shall be determined as specified in 4.6.9. The efficiency of the unit shall be at least 95 percent of the efficiency of the preproduction unit prior to the endurance test.

4.6.6 Overheat controls operation. Satisfactory operation of the overheat controls shall be demonstrated by performing the following operation:

a. The temperature of the airstream shall be increased above 200°F until the limiting device actuates to break the power supply circuit. It shall be shown that the actuation occurs at an airstream temperature of 215° ± 5°F.

b. The temperature of the secondary heat-transfer media shall be increased above the design temperature until the limiting device actuates. It shall be shown that this actuation occurs at a temperature of 325° ± 5°F.

c. Testing shall be conducted to determine compliance with 3.7.3.

4.6.7 Final inspection. After completion of the tests specified in 4.6.5 and 4.6.6, the heater shall be visually inspected to the maximum practical extent to determine compliance with the requirements specified herein.

4.6.8 Accessibility. The following components shall be removed from the heater. The time required and method of removal shall be recorded.

a. Heat generator

b. Heat exchanger

c. Air blower and motor

4.6.9 Capacity. Capacity tests of the heater shall be made prior to any testing and shall be as specified in 3.6.1.2 for operation at 70°F. The capacity of the heater shall be calculated by determining the average temperature of inlet air, outlet air, and the weight of air discharged from the heater.

4.6.10 Mobility. The mobility tests associated with the requirements of MIL-M-8090 in 3.5.6 of this specification shall be performed except as amended in the following paragraphs.

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4.6.10.1 Highways. The heater and ducts shall be towed for a distance of 50 miles at a speed of 20 mph over paved highways. Periodically during this test, the equipment shall be turned through a 360° turn at its minimum turning radius at a speed of 5 mpf.

4.6.10.2 Gravel roads. The heater and ducts shall be towed over graded gravel roads at an average speed of 10 mph (maximum speed 12 1/2 mph) for a distance of 25 miles.

4.6.10.3 Terrain. The heater and ducts shall be towed over rough terrain (unpaved area) at an average speed of 5 mph (maximum speed 7 1/2 mph) for a distance of 5 miles. The rough terrain shall consist of boulder-strewn clay or sod and shall be broken by bumps or depressions averaging at least 4 inches above or below the normal surface dispersed throughout the full length of the course selected.

4.6.10.4 The heater shall then be inspected for misalignment, distortion, and other defects that may have developed during the mobility tests. Following the mobility tests and without intervening reconditioning, the heater shall be checked for proper operation under various temperature and air delivery settings.

4.6.10.5 The tests specified in 4.6.10 through 4.6.10.4 shall be conducted on each class of heater specified in the contract.

4.6.11 Endurance and capacity. The heater shall pass an endurance and capacity test of at least 1,000 hours. The total test time shall be as follows:

a. First period - 250 hours operation at maximum air delivery with a discharge temperature of 200°F, operating a backpressure of 0, 3, 6, 9, and 12 inches water gage (50 hours each). During the final hour of the first period, compliance with 3.6.3 shall be shown by means of a horizontal and vertical temperature traverse (eight measuring points for each traverse) of a symmetrical pattern (which blankets the entire duct) in a cross-sectional plane of a duct, 8 inches in diameter, 10 inches from the heater discharge.

b. Second period - 250 hours operation at maximum air delivery (25 percent) with a discharge temperature of 200°F, operating against back pressures of 0, 3, 6, 9, and 12 inches water gage (50 hours each).

c. Third period - 250 hours operation at maximum air delivery as specified in 4.6.11a, except at a discharge temperature of 150°F.

d. Fourth period - 250 hours operation at minimum air delivery as specified in 4.6.11b, except at a discharge temperature of 150°F.

4.6.11.1 A rated capacity test as specified in 4.6.9 shall be conducted before starting, after each 250-hour interval, and at the end of the endurance tests to demonstrate that no excessive decrease in operating efficiency has occurred. Also, at the completion of the 1,000-hour endurance test, the unit shall be operated as a blower only (with heating system inoperative) and shall deliver at least 45 pm of air at ambient temperature against a static pressure of 0 to 12 inches of water.

4.6.11.2 Compliance with 3.7.4.1 shall be demonstrated at various conditions during the endurance tests.

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4.6.11.3 Airflow temperature limiting device. The airflow temperature limiting device specified in 3.7.4.2 shall be tested for operation during the endurance tests by blocking the airflow or overriding controls in such a manner that outlet temperature is increased.

4.6.12 Operation in tilted position. The complete heater shall be tilted in four different positions, 90° apart, at an angle of 8-1/2° and operated for 30 minutes at maximum output (70° ambient) in each of the various positions to determine compliance with the requirements specified herein.

4.6.13 Noise level. The criteria specified in 3.10 specifies average noise levels for the operator and maximum and average noise levels on a 25-foot radius circle centered on the geometrical center of the unit. The noise levels for the operator are the average levels measured at head level (5 feet 8 inches) in the area where the operator is located. This measurement shall be made by slowly moving the measuring microphone in a circle 2 feet in diameter centered on the most probable position for an operator. The average level for each octave band shall be determined by a simple arithmetic average of the minimum and maximum noise levels (in decibels) found on this circle. The noise levels on the 25-foot radius circle shall be measured at head level (5 feet 8 inches) at 12 positions equally spaced every 30°. The average noise level for each octave band at the operator's position and the average and maximum noise levels in each octave band on the 25-foot circle shall not exceed the values specified in 3.10. Measurements shall be made at maximum airflow and at 200°F outlet temperature. All acoustic measurements shall be made with instrumentation consisting of a sound level meter in conjunction with an octave band analyzer. The instrumentation shall meet the requirements specified in ASA publications S1.4 and S1.11, except the octave band analyzer shall employ the preferred frequencies in ASA publication S1.6. The flat frequency response (C weighting network) of the sound level meter shall be used in making all measurements.

4.6.14 Environmental tests. The heater and all components shall be subjected to the following tests in accordance with the specified procedures of MIL-STD-810 and as specified herein:

4.6.14.1 Low temperature. The low temperature test shall be conducted in accordance with Procedure I of Method 502.1 under the following conditions:

a. Pretest data required: No requirement.

b. Failure criteria: Failure to meet rated output (see 3.6.1.1 and 3.6.1.2) and failure to start (see 3.6.2).

c. Storage temperature and duration: -65°F (-53.9°C) for 24 hours.

d. Lowest operating temperature: -65°F (-53.9°C).

e. Chamber air velocity: Zero.

f. Length of time required for operation and required measurements: Upon completion of the soak period and at an ambient temperature of -65°F, the heater shall be started and operated under various conditions for continuous period of 2 hours. The temperature shall be permitted to increase to -20°F, and the heater shall be tested for 2 hours to assure compliance with 3.6.1.1.

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The temperature shall then be permitted to stabilize at 0°F, and the heater shall be tested for 2 hours to assure compliance with 3.6.1.2 (for operation at 0°F).

4.6.14.2 High temperature. The high temperature test shall be conducted in accordance with Method 501.1 under the following conditions:

a. Procedure number: I

b. Pretest data required: No requirement.

c. Failure criteria: There shall be no evidence of softening of thermoplastic materials or other damage, and the heater shall thereafter meet the other requirements of this specification.

d. Highest operating temperature at which the test item is designed to operate: 125°F (51.8°C) as blower only and 100°F (37.8°C) as heater.

e. Length of time required for operation and required measurements: Upon completion of the 48-hour soak period, the heater temperature shall be lowered to 125°F (51.8°C) and the unit operated as a blower only. The airflow shall be measured at various static pressures from 0 to 12 inches of water. The unit shall be cooled to 100°F and the heater started and checked for proper operation under various air outlet temperatures and flow rates.

4.6.14.3 Rain. The rain test shall be conducted in accordance with Method 506.1 under the following conditions:

a. Procedure number: I

b. Pretest data required: No requirement.

c. Failure criteria: Heater shall be checked for satisfactory operation and material deterioration.

d. Operation during rain: No requirement.

4.6.14.4 Fungus. The fungus test shall be conducted in accordance with Method 508.1 under the following conditions:

a. Pretest data required: No requirement.

b. Failure criteria: Heater shall be checked for satisfactory operation and deterioration of materials.

c. Test period: 28 days.

d. Operation during test: No requirement.

4.6.14.5 Salt fog. The salt fog test shall be conducted in accordance with Method 509.1 under the following conditions:

a. Pretest data required: No requirement.

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b. Failure criteria: Heater shall be checked for satisfactory operation and deterioration of materials.

c. Operation of electrical system during test.

4.6.14.6 Altitude capability. Calculations of airflow rate and static pressure capacity shall be made and submitted to show compliance with specification rated design requirements for operation at 5,000 feet altitude.

4.6.15 Air transportability. The air transportability acceptance tests of MIL-A-8421 shall be performed with the exception of the loading demonstration test. At the conclusion of the flight and taxing load tests, the heater shall be checked for proper operation under various temperature and air delivery settings.

4.6.15.1 Hoisting provisions. The suitability of the tiedown and hoisting provisions shall be demonstrated. The unit shall be hoisted from a single point by attaching four cables of equal length, opposite ends terminating at the common point, at the unit. While statically suspended, the unit shall not tilt at an angle with the horizontal of more than 15°. The unit shall then be hoisted from a single hoisting point with unequal cable lengths that cause the unit to tilt at an angle of 30° with the horizontal. This shall be done once along the lateral axis and once along the traverse axis. In each case, the end nearest the center-of-gravity of the unit shall be tilted downward. The unit shall be checked to determine if the angle with the horizontal increased beyond 30°.

4.6.16 Electromagnetic interference. The heater shall be tested in accordance with and comply with the test requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463. Requirements for Class II B equipment shall apply.

4.6.17 Explosion. Explosion tests on the unit and components shall be performed utilizing a combustible gaseous atmosphere. Determination of a combustible mixture shall be accomplished by routing a portion of the chamber mixture through a relieved test chamber wherein the mixture may be detonated by heating to 375° to 385°F (unless spark ignition is specified). The following tests shall be conducted to determine compliance with the requirements specified herein:

4.6.17.1 Heat exchanger explosion resistance. The heat exchanger shall be placed in a closed circuit containing combustible mixtures of air and a hydrocarbon fuel having an ignition temperature between 350° and 375°F. The heat exchanger discharge air temperature shall be adjustable to 200°F. A sample mixture shall be taken and detonated to prove that a combustible mixture is passing through the heat exchanger. After reaching maximum controlling thermostat temperature, the discharge air thermostat shall be bypassed and the heat exchanger shall be permitted to heat uncontrolled. The maximum obtainable temperature under this condition shall not exceed 375°F.

4.6.17.2 Heat generator explosion containment. The heat generator shall be so placed in an explosion chamber that the heat generator can be filled with a combustible mixture of air and grade JP-4 fuel conforming to MIL-T-5624. The combustible mixture shall be ignited by use of a spark plug. The heat generator shall withstand, without rupture or permanent distortion, a hydrostatic test of 2-1/2 times the maximum internal pressure developed during the explosion test. Calculations indicating a safety factor of 4 based on the maximum explosion pressure will be considered acceptable in lieu of actual hydrostatic testing.

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4.6.17.3 Spark resistance. A combustible mixture of air and grade JP-4 fuel conforming to MIL-T-5624 shall be circulated through the heater. While the mixture is being circulated, a test sample of the fine aggregates shall be introduced into the blower inlet. Test aggregates shall be of the following sizes:

ASTM Sieve No	Percent by Weight of Total Volume
4	70 ± 2
8	20 ± 2
16	10 ± 2

4.6.17.4 Unit explosion tests. The heater shall be tested in accordance with Procedure I of MIL-STD-810. Pretest data is not required. The heater shall be operated normally at maximum output during the test at a chamber temperature of 125°F. An explosion shall constitute failure.

4.6.18 Dielectric. A test voltage of twice the normal voltage (rms) 60 cycle of the circuit to which each circuit is designed to be connected, plus 1,000V rms shall be successively applied for 1 minute between electric circuit and all other electric circuits and metal parts grounded. All circuits, except that under test, shall be connected to ground.

4.6.19 Final capacity. Test sample No 1 shall be subjected to a final capacity test in accordance with 4.6.9.

4.6.20 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to verify conformance to the requirements of Section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each heater shall be preserved and packaged in accordance with the requirements of MIL-E-16298, Method IIa.

5.1.1.1 Ducts. The flexible ducts shall be retracted, wrapped with waterproof barrier material conforming to MIL-B-121, sealed with pressure sensitive tape conforming to PPP-T-60 and stowed in cabinet.

5.1.2 Level B. Each heater shall be preserved and packaged in accordance with the requirements of MIL-P-116, Method I. Plug and seal all openings.

5.1.2.1 Ducts. The flexible ducts shall be retracted and stowed in cabinet.

5.1.3 Level C. Heaters shall be preserved and packaged in accordance with the manufacturer's standard commercial practice which shall ensure complete protection against deterioration and damage during shipment to first destination.

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5.2 Packing

5.2.1 Level A. Heaters preserved and packaged as specified in paragraph 5.1.1 shall be boxed in accordance with MIL-B-26195, Type II, Style A, Class 1, Full Panel with cleated plywood superstructure by bolting. Heater shall be mounted to container base by using angle iron brackets of the type specified in Figure 1. Mounting brackets shall be bolted through load bearing members or the skids of the base. Anchoring of the towbar shall be accomplished in accordance with MIL-STD-1186.

5.2.2 Level B. Heaters preserved and packaged as specified in paragraph 5.1.2 shall be shipped mobile.

5.2.3 Level C. Heaters preserved and packaged as specified in paragraph 5.1.3 shall be shipped mobile.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

5.3.1 Special marking. "DO NOT FORKLIFT" in capital letters not less than 1 inch in height shall be stenciled on each side of the heater cabinets.

6. NOTES

6.1 Intended use. The HDU-13/M heater is intended for use in hangars and nose docks for fuel cell repair operations and for the heating of aircraft compartments by delivery of heated air at moderate pressures through an airborne duct system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Location and conditions for preproduction testing (see 4.4).

c. Levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent.

$$MTBF = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

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The 2.3 = $\frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an expontential (2 degrees of freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.5 Definitive specification part number. The specification part number is definitive part number which will be formulated to identify each item covered by this specification. The part number will be formulated by selection from the requirement options available in this specification as follows:

Definitive Specification Part Number M27505 - X

Military Specification Number

Class Designator (see 6.5.1)

6.5.1 Class designator. A one-position field used to designate the required class of heater (see Table III).

TABLE III

Class Designator	Remarks
I	Equipped with a 220/440v, 3 phase, 60 Hz electric motor
II	Equipped with a 220v, 3 phase, 400 Hz electric motor

6.6 International standardization agreements. Certain provisions of this specification (identified by paragraph number or similar manner, if appropriate) are the subject of international standardization agreement (insert the ABCA or ASCC standard number, the NATO, STANAG, NETR or NEPR number, or other appropriate documentary reference). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

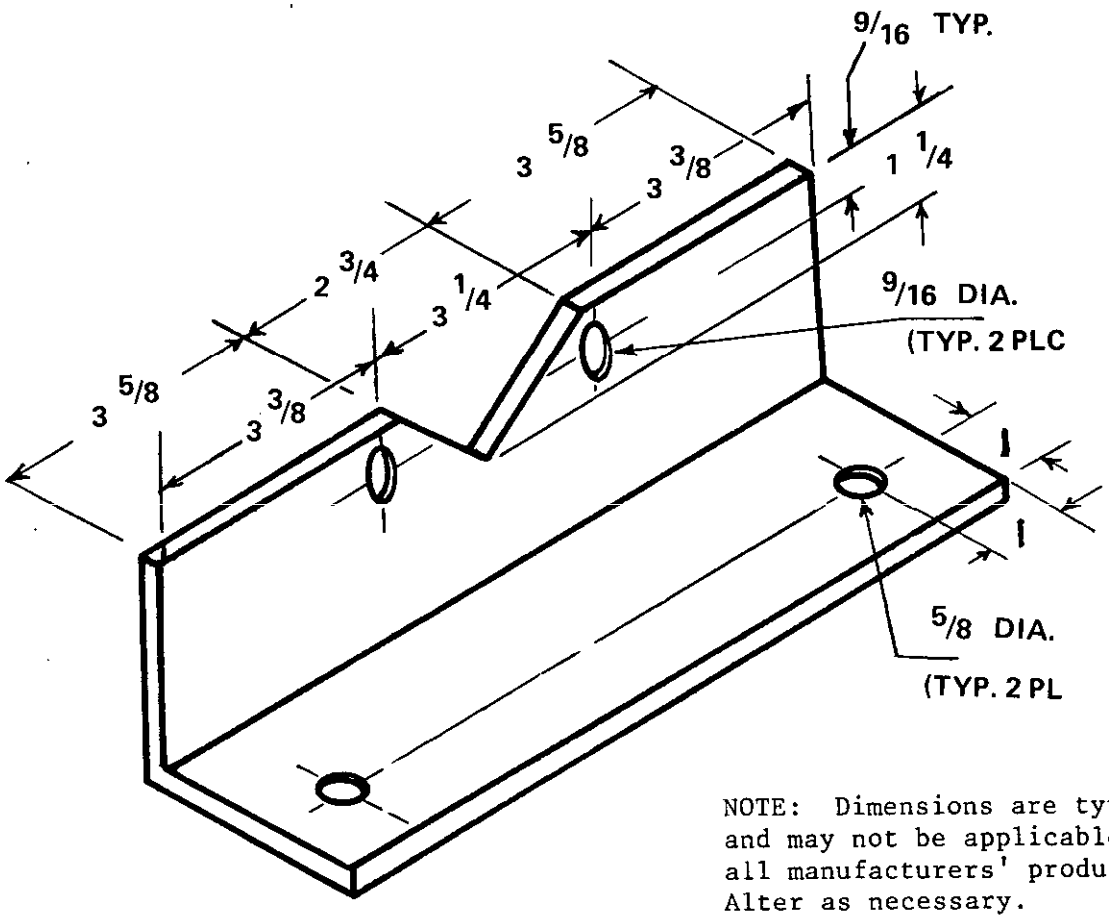
Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

Project No:
4520-F202

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MOUNTING BRACKET
AEROSPACE GROUND EQUIPMENT
(STEEL, 3 X 3 X 1/4 X 10)



NOTE: Dimensions are typical and may not be applicable to all manufacturers' products. Alter as necessary.

FIGURE 1

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